

MongoDB

What is MongoDB?

Definition:

MongoDB is a **NoSQL, document-oriented** database that stores data in the form of **documents** instead of rows and columns.

- SQL databases store data in **tables**.
- MongoDB stores data in **collections**.
- Each record is called a **document**.

Example

SQL Database

Student Table

ID	Name	Age
1	Rahul	20
2	Priya	21

Here the data is stored in **rows**.

MongoDB Database

Student Collection

```
{
  "_id": 1,
  "name": "Rahul",
  "age": 20
}
```

```
{  
  "_id": 2,  
  "name": "Priya",  
  "age": 21  
}
```

Here, each student is a **document**.

What is NoSQL?

NoSQL = Not Only SQL

It means the database **does not use tables with fixed columns** like traditional relational databases.

MongoDB stores data in **JSON-like documents** (internally in BSON).

Why was NoSQL introduced?

Suppose you have a Student table.

SQL

ID Name Age

1 Rahul 20

Now you want to add **Email**.

ID Name Age Email

In SQL, you usually need to alter the table structure:

ALTER TABLE Student

ADD Email VARCHAR(100);

MongoDB

Old document

```
{  
  "name": "Rahul",  
  "age": 20  
}
```

New document

```
{  
  "name": "Priya",  
  "age": 21,  
  "email": "priya@gmail.com"  
}
```

No need to change the collection structure.

This is called a **flexible schema**.

What is a Document?

A **document** is a single record in MongoDB.

Example

```
{  
  "name": "Rahul",  
  "age": 20,  
  "city": "Chennai"  
}
```

Here,

- name → field
- age → field

- city → field

The whole object is a **document**.

What is a Collection?

A **collection** is a group of documents.

Example

Student Collection

```
{  
  "name": "Rahul",  
  "age": 20  
}
```

```
{  
  "name": "Priya",  
  "age": 21  
}
```

```
{  
  "name": "Arun",  
  "age": 19  
}
```

Just like a **table** in SQL.

What is a Database?

A **database** contains one or more collections.

Example

College Database

|

|— Student Collection

|— Faculty Collection

|— Course Collection

└— Department Collection

Why is MongoDB Popular?

1. Flexible Schema

You can store different fields in different documents.

Example

```
{  
  "name": "Rahul",  
  "age": 20  
}  
  
{  
  "name": "Priya",  
  "age": 21,  
  "email": "priya@gmail.com"  
}
```

Both documents can exist in the same collection.

2. Fast

MongoDB provides fast read and write operations because of efficient indexing and its document model.

3. Easy to Scale

If your application grows from 1,000 users to 10 million users, MongoDB supports **horizontal scaling** using **sharding**.

4. JSON-like Data

Documents are easy for developers to read and map directly to objects in many programming languages.

MongoDB vs SQL

SQL	MongoDB
Database	Database
Table	Collection
Row	Document
Column	Field
Primary Key	_id
Fixed schema	Flexible schema
SQL queries	MongoDB queries

Advantages of MongoDB

- Flexible schema
 - High performance
 - Easy to scale
 - Supports large amounts of data
 - Stores nested (embedded) documents
 - Good for web and mobile applications
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Disadvantages of MongoDB

- More difficult to perform complex joins than in SQL
 - Can use more storage because field names are stored with the data
 - Flexible schemas require developers to enforce consistency in many applications
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Frequently Asked Interview Questions

Q1. What is MongoDB?

Answer:

MongoDB is a NoSQL, document-oriented database that stores data in JSON-like documents (BSON) instead of rows and columns. It provides a flexible schema, high performance, and easy scalability.

Q2. Why is MongoDB called a NoSQL database?

Answer:

MongoDB is called a NoSQL database because it does not store data in relational tables. Instead, it stores data as collections of documents with a flexible schema.

Q3. What is a document?

Answer:

A document is a single record in MongoDB. It stores data as key-value pairs, similar to a JSON object.

Example:

```
{  
  "name": "Rahul",  
  "age": 20  
}
```

Q4. What is a collection?

Answer:

A collection is a group of related documents, similar to a table in a relational database.

Q5. What is the difference between a document and a collection?

Document	Collection
Single record	Group of records
Similar to a row	Similar to a table